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Distributed Finite-Time Control of Islanded Microgrid for Ancillary Services Provision

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Abstract: This paper presents a hierarchical cyber-physical multi-agent model for an AC microgrid (MG). A new distributed finite-time secondary controller is designed in the provision of ancillary services, such as voltage and frequency synchronization and active and reactive power regulation. The control algorithm developed is fully distributed and intended to restore the system voltage and frequency to their nominal value finite time. The existing distributed controllers achieve a consensus in an infinite-time horizon, whereas the proposed control provides a quick convergence to a consensus even in the face of disturbances and restores the voltage and frequency in less than 0.25 s. For accurate active and reactive power regulation, a distributed control algorithm is proposed that corrects the power mismatch among neighboring distributed generator units and eventually for the entire MG network. A Lyapunov analysis is used to establish the upper limit on the convergence time. The proposed control scheme is evaluated and compared with the previously reported asymptotic control technique based on a neighborhood tracking error using time-domain simulations under load variation and communication constraints. The controller withstands a communication link delay up to 2 s with a small deviation and settles down to the nominal values within 0.4 s. Further, the convergence analysis shows that the performance of the proposed algorithm depends exclusively on the controller parameters and communication network connectivity, not on the line and load parameters of the AC MG. The proposed controller enables the plug-and-play capability of the AC MG and effectively reduces the load change-induced disturbance.



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1. Introduction

The interconnected configuration of an MG provides better reliability during faulty operating conditions compared to radial topology. Being more reliable than the conventional power system, the MG can support an islanded and grid-connected mode of operation. In a grid-tied operation, the MG depends on the utility grid for the ancillary services, whereas the control in an islanded operation is more demanding because of the low inertia and no rotating mass. The operating voltage and frequency of the islanded MG have to be appropriately synchronized for achieving a stable operation. Because of the heterogeneous nature of renewable energy sources, the power electronics inverters are used to interface the DGs to the main grid. Thus, the control of these inverters in a synchronized manner is of paramount importance [1].

A three-layered control structure is usually adopted to regulate the operation of the microgrid. For voltage and frequency synchronization and proper reactive and active power sharing, droop-based primary control is realized at each DG terminal. Regardless of the simple implementation and distributed structures, the droop control leads to a voltage and frequency divergence and poor power regulation. To address the power

regulation problem, a modified droop-based control is proposed in [2], but in this work, active power sharing is not considered. Moreover, reactive power sharing cannot be achieved with a highly non-uniform line impedance. An adaptive droop-based scheme is reported in [3] for reactive power regulation. However, reactive power regulation and frequency synchronization are coupled together, so finite-time frequency restoration cannot be achieved. The secondary level of control is implemented as the supervisory control to synchronize the voltage and frequency by the updation of the reference in the primary control layer. Several secondary control algorithms based on a centralized control structure are reported in the literature [4]. The central controller has several drawbacks, such as global information collection, the processing of huge data, and a complex bidirectional communication network, which raises reliability and flexibility concerns along with the risk of one-point failure. The tertiary control level works on the bidirectional power flow and cost optimization and load dispatch. During the past decade, the secondary layer of control has grown from centralized to decentralized to distributed control, improving the scalability and flexibility and providing a simple communication network. Most of the distributed control methods are designed on the basis of a consensus algorithm that operates on an infinite-time horizon and reaches a consensus asymptotically [4–6].

A distributed voltage regulation scheme based on a consensus algorithm is presented in [7], where the frequency synchronization and active power regulation are not considered. Only the voltage and reactive power regulation are considered in [8]. A feedback linearization-based voltage synchronization approach is proposed in [4], but it has not considered the power sharing and frequency control. Further, ref. [9] considers only the frequency regulation, without taking care of the voltage control or power sharing. Voltage and frequency synchronization are considered in [5], but reactive power regulation is overlooked. A distributed controller for the voltage, frequency, and accurate reactive power regulation is presented in [6], but it needs the bidirectional communication network for an information exchange among DG units for an entire MG system. Moreover, it does not provide any stability analysis for the proposed system. A distributed control paradigm is proposed in [10] by replacing the primary droop control level with regulators for the voltage and power. However, the absence of a droop controller makes the system communication dependent; hence, the controller performance deteriorates with communication constraints.

The continuous operation of sensitive loads in an MG requires the standard voltage and frequency regulation. The secondary voltage and frequency regulation methods achieve a consensus over an infinite-time horizon and are asymptotically stable. To obtain a faster convergence, finite-time restoration approaches are reported in the literature [11–13]. The fundamental task in the MG network modeled as a multi-agent system is the consensus, and numerous results were obtained to achieve a finite-time consensus. The finite-time controller improves the system dynamic performance, stability under perturbation, and efficiency along with the definite convergence time [14]. The work in [11] provides a finite-time frequency restoration method without considering the voltage and reactive power control. Further, ref. [15] considers both the voltage and frequency restoration but does not include the reactive power sharing. A virtual impedance-based finite-time approach is proposed in [12] which is further improved by combining information sharing and an adaptive virtual impedance approach for the reactive power allocation in [13]. For a fixed-time consensus, some related recent control algorithms are reviewed in detail in [16] and provide a set of finite-time approaches for the cooperative control of an MG. A distribution grid with a high DG penetration has power management-related problems that are addressed using a smart transformer for an AC MG in [17,18]. A detailed review on various distributed controllers for microgrid synchronization is presented in [19]. A noise resilient distributed controller is presented in [20] for the voltage and frequency synchronization under noisy communication links. A distributed controller with a bounded input constraint is designed in [21] for practical applications and fast-changing operating conditions. The authors in [22,23] present a multi-objective approach based on modified sliding mode control for voltage frequency restoration and optimal power sharing, where

the controller does not depend on the leader–follower-based consensus. However, the analysis does not consider the communication constraints.

Research Gap, Motivation, and Contributions

The existing secondary control schemes for an islanded MG are either asymptotic in nature or, if time bounded, then do not provide the upper limit on the convergence time. Moreover, the stability analysis is not addressed thoroughly with proof. Further, few papers address only the voltage and reactive power sharing or frequency and active power regulation. The communication constraints which are an inherent part of the secondary control layer are not considered under the verification of the controllers. Motivated by the above-mentioned research gaps, in this work, we have proposed a finite-time control for the multi-objective ancillary service provision, i.e., the voltage and frequency restoration, along with accurate power sharing which was not addressed with an upper limit on the synchronization time. Moreover, the Lyapunov stability analysis is described in detail which was not presented in the existing literature.

In this paper, a new finite-time secondary restoration algorithm for the voltage and frequency synchronization and an asymptotic control scheme for accurate power sharing for an AC MG are presented. The main contributions of the proposed controller are as follows:

- Unlike the existing distributed algorithms, which converge over the infinite-time horizon, we develop a finite-time restoration algorithm for both the voltage and frequency synchronization of the stand-alone MG.
- The upper limit on the voltage restoration time T_V is provided which only depends on the design parameters of the controller and not on the system states. This allows the off-line estimation and tuning of the synchronization time in accordance with the operating condition and user demand.
- Separate asymptotic controllers are designed to work on a slower time frame for an accurate active and reactive power allocation among the DGs in accordance with their power ratings.
- The proposed restoration scheme is completely distributed in nature; also, the DG only requires the local and neighbor's information exchanged over a sparse communication network. The presented controller enables the plug-and-play feature of the AC MG.

The rest of this paper is organized as follows. Section 2 introduces the cyber-physical AC MG framework. The proposed finite-time restoration scheme is given in Section 3. Finally, validation using the simulation cases is given in Section 4. Section 5 concludes the paper.

2. Cyber-Physical Framework of AC Microgrid

The microgrid system is modeled as a typical two-tier hierarchical multi-agent system (MAS), as shown in Figure 1a. The first tier is a physical layer comprising the electrical system elements, such as DG units, electrical loads, and transmission lines. The second tier is cyber communication network for information exchange among the DG units, which facilitates the use of different restoration paradigms to achieve consensus among the DGs on the quantity of interest. The cyber framework is designed by mapping DG unit in an MG system as an agent of MAS. The obtained cyber network is a cyber framework where DGs share necessary information with their neighboring DG unit. The proposed controller inputs are updated on the basis of received information from neighboring DGs units. The cyber layer can be formed by a weighted undirected connected graph. Details on the graph theory can be found in [4].

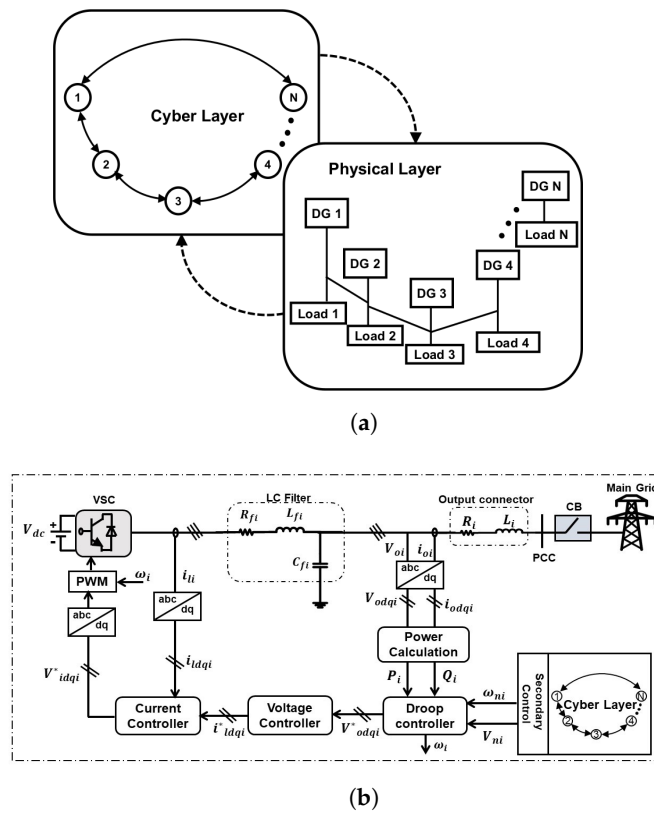


Figure 1. (a) Cyber-physical microgrid system. (b) Dynamical model of DG unit.

2.1. Preface on Graph Theory

It can be represented as $G = (v_G, E_G, A_G)$ with $v_G = \{1, 2, \dots, N\}$ set of nodes, $E_G \subset v_G \times v_G$ is a set of edges (communication links), and $A_G = [a_{ij}] \in R^{N \times N}$ is associated adjacency matrix. a_{ij} is the weight of the edge (i, j) , if $a_{ij} = 1$, i th node receives information from j th node, otherwise $a_{ij} = 0$. The Laplacian matrix of graph G is $\mathcal{L}_G = D_G - A_G$, where D_G is the in-degree matrix defined as $D_G = \text{diag}\{d_i\}$, and diagonal matrix with $d_i = \sum_{j \in N_i} a_{ij}$, i.e., the number of incoming links at node i . $\mathcal{L}_G$ has all row sum equal to zero, i.e., $\mathcal{L}_G \mathbf{1}_N = 0$, where $\mathbf{1}_N$ is the vector of all ones of length N . The eigenvalues of $\mathcal{L}_G$ have at least one zero entry with all other positive real parts and determine the global dynamics of the system.

2.2. Preface on Primary Control Layer

Figure 1b shows the control structure for the i th inverter-based DG unit. It consists of the droop-based primary control layer and a secondary control block. The latter comprises voltage, frequency, and power-sharing controllers. A comprehensive model for inverter-based DG is given in [10]. The V-f droop characteristics can be expressed as [4],

$$V_{odi}^* = V_{ni} - k_{Qi}^V Q_i \quad (1)$$

$$\omega_i = \omega_{ni} - k_{Pi}^\omega P_i \quad (2)$$

where V_{odi}^* and ω_i are output voltage magnitude and operating angular frequency, V_{ni} and ω_{ni} are the reference voltage and frequency for primary controller, which are updated using secondary control layer for synchronization. k_{Qi}^V and k_{Pi}^ω are voltage and frequency droop coefficients for i th DG, respectively. The secondary controller provides updated control input for voltage and frequency deviation resulting due to the primary droop control layer. The proposed distributed secondary finite-time control scheme achieves these ancillary services effectively and provides correct active power sharing among DG units according to their ratings.

3. Distributed Finite-Time Restoration

In this section, the distributed finite-time voltage and frequency restoration schemes are proposed.

3.1. Voltage and Reactive Power Regulation

For voltage synchronization and accurate reactive power sharing among DG units, the control system is designed by differentiating droop equation given in (1) as follows.

$$\begin{aligned}\dot{V}_{ni} &= \dot{V}_{odi}^* + k_{Qi}^V \dot{Q}_i = u_i^V + u_i^Q, \\ V_{ni} &= \int (u_i^V + u_i^Q) dt\end{aligned}\quad (3)$$

where $u_i^V = \dot{V}_{odi}^*$ and $u_i^Q = k_{Qi}^V \dot{Q}_i$ are the auxiliary control variables for voltage and reactive power regulation. The controller for voltage and reactive power regulation is given by (3). We take up the design of auxiliary voltage control first and design the auxiliary reactive power control later. u_i^V can be designed based on the multi-agent consensus control theory and neighborhood tracking error, and its general form can be written as follows [14]:

$$\begin{aligned}u_i^V &= \gamma^V \operatorname{sgn} \left(\sum_{j \in \mathcal{N}_i} a_{ij} (V_{odj} - V_{odi}) \right) - \eta^V \operatorname{sgn}(V_{odi} - V_{nom}) - \beta^V \sum_{j \in \mathcal{N}_i} \mathcal{L}_G V_{odj} - \mathcal{V}_i^V \\ &\quad + g_1^V \left(\sum_{j \in \mathcal{N}_i} a_{ij} (V_{odj} - V_{odi}) \right)^{\frac{\mu_1}{\mu_2}} + g_2^V \left(\sum_{j \in \mathcal{N}_i} a_{ij} (V_{odj} - V_{odi}) \right)^{\frac{\mu_3}{\mu_4}}, \\ \dot{\mathcal{V}}_i^V &= \alpha^V \beta^V \sum_{j \in \mathcal{N}_i} \mathcal{L}_G V_{odj}\end{aligned}\quad (4)$$

where $\gamma^V, \eta^V, \alpha^V, \beta^V, g_1^V$, and g_2^V are positive coefficients used to tune the performance of the controller and $\operatorname{sgn}(x)$ is the signum function. $\mathcal{L}_G$ is the Laplacian matrix of the weighted graph G and μ_1, μ_2, μ_3 , and μ_4 are all positive odd integers, satisfying $\mu_1 > \mu_2$ and $\mu_3 < \mu_4$.

Remark 1. It is assumed that the nominal values of voltage and frequency are available with DG1 (leader node) and shared using the communication network for implementing the secondary controller. The voltage reference term V_{nom} in (4) and the angular frequency term ω_{nom} in (26) are the received information over the communication digraph.

Theorem 1. The auxiliary controller given in (4) restores the voltage magnitude of DGs unit to

$$\text{their nominal value in definite convergence time } T_V \leq \frac{1}{|\lambda_2(\mathcal{L}_G)|} \left(\frac{N^{\frac{\mu_1 - \mu_2}{2\mu_2}} \mu_2}{g_1^V (\mu_1 - \mu_2)} + \frac{\mu_4}{g_2^V (\mu_4 - \mu_3)} \right).$$

Proof. To demonstrate the stability of the presented voltage restoration scheme, we consider a Lyapunov candidate function as follows,

$$\mathcal{V}(\mathbf{V}) = \frac{1}{4} \sum_{i=1}^N \sum_{j=1}^N a_{ij} (V_{odi} - V_{odj})^2 = \frac{1}{2} \mathbf{V}^T \mathcal{L}_G \mathbf{V} \quad (5)$$

where $\mathbf{V} = [V_{od1}, V_{od2}, \dots, V_{odN}]^T$. Because G is a connected graph, zero will be a simple eigenvalue of Laplacian $\mathcal{L}_G$. The partial derivative of the considered positive Lyapunov function corresponding to each state variable is given as

$$\frac{\partial \mathcal{V}(\mathbf{V})}{\partial V_{odi}} = - \sum_{j=1}^N a_{ij} (V_{odj} - V_{odi}) \quad (6)$$

The time derivative of $\mathcal{V}(\mathbf{V})$ is

$$\dot{\mathcal{V}}(\mathbf{V}) = \sum_{j=1}^N \frac{\partial \mathcal{V}(\mathbf{V})}{\partial V_{odi}} \dot{V}_{odi} = - \sum_{j=1}^N a_{ij} (V_{odj} - V_{odi}) u_i^V \quad (7)$$

Substituting (4) into (7), and using $x \operatorname{sgn}(x) = |x|$, and $\delta_{V_{odj} j \in \mathcal{N}_i} = \beta^V \sum_{j \in \mathcal{N}_i} \mathcal{L}_A V_{odj} + \mathcal{V}_i^V$, we obtain

$$\begin{aligned} \dot{\mathcal{V}}(\mathbf{V}) = & -\gamma^V \sum_{i=1}^N \left| \sum_{j=1}^N a_{ij} (V_{odj} - V_{odi}) \right| + \eta^V \operatorname{sgn}(V_{odi} - V_{nom}) \sum_{i=1}^N \sum_{j=1}^N a_{ij} (V_{odj} - V_{odi}) \\ & + \sum_{i=1}^N \sum_{j=1}^N a_{ij} (V_{odj} - V_{odi}) \delta_{V_{odj} j \in \mathcal{N}_i} - g_1^V \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij} (V_{odj} - V_{odi}) \right)^{\frac{\mu_1 + \mu_2}{\mu_2}} \\ & - g_2^V \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij} (V_{odj} - V_{odi}) \right)^{\frac{\mu_3 + \mu_4}{\mu_4}} \end{aligned} \quad (8)$$

Defining $d^V = \max\{\eta^V + \delta_{V_{od1}}, \dots, \eta^V + \delta_{V_{odN}}\}$, we can rewrite (8) as

$$\begin{aligned} \dot{\mathcal{V}}(\mathbf{V}) = & -(\gamma^V - d^V) \sum_{i=1}^N \left| \sum_{j=1}^N a_{ij} (V_{odj} - V_{odi}) \right| - g_1^V \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij} (V_{odj} - V_{odi}) \right)^{\frac{\mu_1 + \mu_2}{\mu_2}} \\ & - g_2^V \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij} (V_{odj} - V_{odi}) \right)^{\frac{\mu_3 + \mu_4}{\mu_4}} \end{aligned} \quad (9)$$

If $\gamma^V > d^V$, then (9) can be rewritten as

$$\dot{\mathcal{V}}(\mathbf{V}) \leq -g_1^V \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij} (V_{odj} - V_{odi}) \right)^{\frac{\mu_1 + \mu_2}{\mu_2}} - g_2^V \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij} (V_{odj} - V_{odi}) \right)^{\frac{\mu_3 + \mu_4}{\mu_4}} \quad (10)$$

Lemma 1. [24]: Let $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_N \geq 0$ and $0 < P \leq 1$. Then, $-\sum_{i=1}^N \varepsilon_i^P \leq -\left(\sum_{i=1}^N \varepsilon_i\right)^P$.

Lemma 2. [24]: Let $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_N \geq 0$ and $P > 1$. Then, $-\sum_{i=1}^N \varepsilon_i^P \leq -N^{1-P} \left(\sum_{i=1}^N \varepsilon_i\right)^P$.

Lemma 3. [24]: If $\mathcal{L}_G$ is the Laplacian matrix of undirected graph G with 0 as an eigenvalue and $\mathbf{1}_N$ as related eigenvector, then the second smallest eigenvalue of $\mathcal{L}_G$ is $\min_{x \neq 0, \mathbf{1}_N^T x = 0} \frac{x^T \mathcal{L}_G x}{x^T x}$, and if $\mathbf{1}_N^T x = 0$, then $x^T \mathcal{L}_G x \geq \lambda_2(\mathcal{L}_G) x^T x$.

With $\frac{\mu_1 + \mu_2}{2\mu_2} \in (0, \infty)$ and $\frac{\mu_3 + \mu_4}{2\mu_4} \in (0, 1)$, using Lemmas 1 and 2, we can modify (10) as

$$\dot{\mathcal{V}}(\mathbf{V}) \leq -g_1^V N^{\frac{\mu_2 - \mu_1}{2\mu_2}} \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij} (V_{odj} - V_{odi})^2 \right)^{\frac{\mu_1 + \mu_2}{2\mu_2}} - g_2^V \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij} (V_{odj} - V_{odi})^2 \right)^{\frac{\mu_3 + \mu_4}{2\mu_4}} \quad (11)$$

Using Lemma 3, we can write

$$\lambda_2(\mathcal{L}_G) = a(G) = \min_{\mathbf{V} \neq 0, 1_N^T \mathbf{V} = 0} \frac{\mathbf{V}^T \mathcal{L}_G \mathbf{V}}{\mathbf{V}^T \mathbf{V}} \quad (12)$$

If $1_N^T \mathbf{V} = 0$, then $\mathbf{V}^T \mathcal{L}_G \mathbf{V} \geq \lambda_2(\mathcal{L}_G) \mathbf{V}^T \mathbf{V}$ or $\frac{\mathbf{V}^T \mathcal{L}_G \mathbf{V}}{\mathbf{V}^T \mathbf{V}} \geq \lambda_2(\mathcal{L}_G)$, then for $\mathcal{V}(\mathbf{V}) \neq 0$

$$\sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij} (V_{odj} - V_{odi}) \right)^2 \geq \lambda_2(\mathcal{L}_G) \mathcal{V}(\mathbf{V}) \quad (13)$$

Substituting (13) in (11), we can write

$$\dot{\mathcal{V}}(\mathbf{V}) \leq -g_1^V N^{\frac{\mu_2 - \mu_1}{2\mu_2}} (2|\lambda_2(\mathcal{L}_G)| \mathcal{V}(\mathbf{V}))^{\frac{\mu_1 + \mu_2}{2\mu_2}} - g_2^V (2|\lambda_2(\mathcal{L}_G)| \mathcal{V}(\mathbf{V}))^{\frac{\mu_3 + \mu_4}{2\mu_4}} \quad (14)$$

Define $y = \sqrt{2|\lambda_2(\mathcal{L}_G)| \mathcal{V}(\mathbf{V})}$, we can rewrite (14) as

$$\dot{\mathcal{V}}(\mathbf{V}) \leq -g_1^V N^{\frac{\mu_2 - \mu_1}{2\mu_2}} y^{\frac{\mu_1 + \mu_2}{\mu_2}} - g_2^V y^{\frac{\mu_3 + \mu_4}{\mu_4}} \quad (15)$$

Time derivative of y is given as

$$\dot{y} = \frac{2|\lambda_2(\mathcal{L}_G)| \dot{\mathcal{V}}(\mathbf{V})}{2\sqrt{2|\lambda_2(\mathcal{L}_G)| \mathcal{V}(\mathbf{V})}} = \frac{\lambda_2(\mathcal{L}_G) \dot{\mathcal{V}}(\mathbf{V})}{y} \quad (16)$$

Substitute $\dot{\mathcal{V}}(\mathbf{V})$ from (15) in (16), we obtain

$$\dot{y} = \frac{\lambda_2(\mathcal{L}_G)}{y} \left(-g_1^V N^{\frac{\mu_2 - \mu_1}{2\mu_2}} y^{\frac{\mu_1 + \mu_2}{\mu_2}} - g_2^V y^{\frac{\mu_3 + \mu_4}{\mu_4}} \right) \quad (17)$$

Using Lemma 4.1 in [24], (17) represents a global finite-time stable system, where the convergence time is given as follows:

$$T_V \leq \frac{1}{|\lambda_2(\mathcal{L}_G)|} \left(\frac{N^{\frac{\mu_1 - \mu_2}{2\mu_2}} \mu_2}{g_1^V (\mu_1 - \mu_2)} + \frac{\mu_4}{g_2^V (\mu_4 - \mu_3)} \right) \quad (18)$$

T_V is the restoration time for voltage. $\square$

Remark 2. From (18), it is evident that the upper limit of the convergence time T_V exclusively relies on the controller design parameters $\mu_1, \mu_2, \mu_3, \mu_4, g_1^V, g_2^V, N$, and communication network integration, i.e., second smallest eigenvalue of Laplacian matrix ($\mathcal{L}_G$). This facilitates tuning of convergence time using controller gains only.

Next, we design the auxiliary reactive power control u_i^Q as follows:

$$u_i^Q = -c_Q e_{Qi}^V = k_{Qi}^V \dot{Q}_i \quad (19)$$

where c_Q is the controller gain, and e_{Qi}^V is the neighborhood tracking error for reactive power allocation and is given as

$$e_{Qi}^V = \sum_{j \in \mathcal{N}_i} a_{ij} (k_{Qi}^V Q_i - k_{Qj}^V Q_j) \quad (20)$$

Equation (20) is modified to obtain a more compact matrix form as follows.

$$k_Q^V \dot{Q} = u^Q = -c_Q e_Q^V = -c_Q \mathcal{L}_G k_Q^V Q \quad (21)$$

$$\text{where } k_Q^V \dot{Q} = [k_{Q1}^V \dot{Q}_1, k_{Q2}^V \dot{Q}_2, \dots, k_{QN}^V \dot{Q}_N]^T, u^Q = [u_1^Q, u_2^Q, \dots, u_N^Q]^T, \\ e_Q^V = [e_{Q1}^V, e_{Q2}^V, \dots, e_{QN}^V]^T, k_Q^V Q = [k_{Q1}^V Q_1, k_{Q2}^V Q_2, \dots, k_{QN}^V Q_N]^T.$$

Now, consider a Lyapunov function,

$$V(Q) = \frac{1}{2} e_Q^{V^T} R e_Q^V \quad (22)$$

where $R = R^T$ and $R > 0$ is a positive semidefinite matrix. Time derivative of (22) gives

$$\dot{V}(Q) = e_Q^{V^T} R \mathcal{L}_G u^Q \quad (23)$$

Define $M \equiv \mathcal{L}_G$, and using (21) in (24), we can write

$$\dot{V}(Q) = -c_Q e_Q^{V^T} R M e_Q^V = -\frac{1}{2} c_Q e_Q^{V^T} (R M + M^T R) e_Q^V \quad (24)$$

Remark 3. From [25], $R M + M^T R$ is a positive definite matrix; therefore, $\dot{V}(Q)$ will be negative. Thus, the proposed controller is asymptotically stable. Hence, the controller in (19) shares the reactive power according to their droop coefficients in a longer time frame, once the secondary controller is activated. Figure 2 shows the block diagram for proposed voltage and reactive power-sharing algorithm.

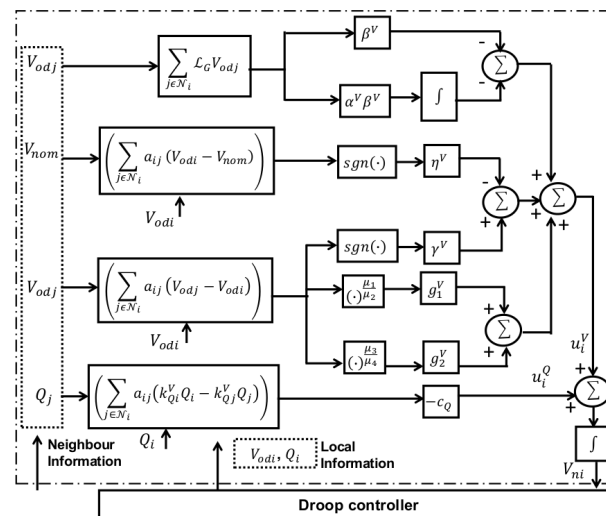


Figure 2. Block diagram of proposed distributed finite-time secondary voltage and reactive power regulation scheme.

3.2. Frequency and Active Power Regulation

For frequency and active power regulation problem in stand-alone MG, the control input can be designed by differentiating (2)

$$\dot{\omega}_{ni} = \dot{\omega}_i + k_{P_i}^\omega \dot{P}_i = u_i^\omega + u_i^P, \omega_{ni} = \int (u_i^\omega + u_i^P) dt \quad (25)$$

where $u_i^\omega = \dot{\omega}_i$ and $u_i^P = k_{P_i}^\omega \dot{P}_i$ are the ancillary controller inputs for frequency synchronization and active power sharing, respectively. The auxiliary control input for frequency u_i^ω can be designed as follows:

$$\begin{aligned}
u_i^\omega &= \gamma^\omega \operatorname{sign} \left(\sum_{j \in \mathcal{N}_i} a_{ij} (\omega_j - \omega_i) \right) - \eta^\omega \operatorname{sign} (\omega_i - \omega_{nom}) - \beta^\omega \sum_{j \in \mathcal{N}_i} \mathcal{L}_G \omega_j - \mathcal{V}_i^\omega \\
&\quad + g_1^\omega \left(\sum_{j \in \mathcal{N}_i} a_{ij} (\omega_j - \omega_i) \right)^{\frac{\mu_1}{\mu_2}} + g_2^\omega \left(\sum_{j \in \mathcal{N}_i} a_{ij} (\omega_j - \omega_i) \right)^{\frac{\mu_3}{\mu_4}}, \\
\dot{\mathcal{V}}_i^\omega &= \alpha^\omega \beta^\omega \sum_{j \in \mathcal{N}_i} \mathcal{L}_G \omega_{odj}
\end{aligned} \tag{26}$$

where γ^ω , η^ω , α^ω , β^ω , g_1^ω , and g_2^ω are positive coefficients used to tune the performance of the controller.

The auxiliary control input for active power u_i^P can be given as

$$u_i^P = -c_P e_{Pi}^\omega = k_{Pi}^\omega \dot{P}_i \tag{27}$$

where c_P is the controller gain, and e_{Pi}^ω is the neighborhood tracking error for active power sharing and can be given as

$$e_{Pi}^\omega = \sum_{j \in \mathcal{N}_i} a_{ij} (k_{Pi}^\omega P_i - k_{Pj}^\omega P_j) \tag{28}$$

Theorem 2. The ancillary control input presented in (26) and (30) provides the finite-time frequency restoration with convergence time $T_F \leq \frac{1}{|\lambda_2(\mathcal{L}_G)|} \left(\frac{N^{\frac{\mu_1 - \mu_2}{2\mu_2}} \mu_2}{g_1^\omega (\mu_1 - \mu_2)} + \frac{\mu_4}{g_2^\omega (\mu_4 - \mu_3)} \right)$ and active power sharing in longer time frame, respectively.

Proof. To demonstrate the stability of the presented frequency restoration scheme, we consider a Lyapunov candidate function as follows,

$$V(\omega) = \frac{1}{4} \sum_{i=1}^N \sum_{j=1}^N a_{ij} (\omega_i - \omega_j)^2 = \frac{1}{2} \omega^T \mathcal{L}_G \omega \tag{29}$$

where $\omega = [\omega_1, \omega_2, \dots, \omega_N]^T$. Because G is a connected graph, zero will be a simple eigenvalue of Laplacian $\mathcal{L}_G$, and the partial derivative of the considered positive Lyapunov function with respect to each state variable can be given as

$$\frac{\partial V(\omega)}{\partial \omega_i} = - \sum_{j=1}^N a_{ij} (\omega_j - \omega_i) \tag{30}$$

The derivative of $V(\omega)$ versus time is

$$\dot{V}(\omega) = \sum_{i=1}^N \frac{\partial V(\omega)}{\partial \omega_i} \dot{\omega}_i = \sum_{i=1}^N \frac{\partial V(\omega)}{\partial \omega_i} u_i^\omega = - \sum_{i=1}^N \sum_{j=1}^N a_{ij} (\omega_j - \omega_i) u_i^\omega \tag{31}$$

Substitute (26) into (31),

$$\begin{aligned}\dot{V}(\omega) = & - \sum_{i=1}^N \sum_{j=1}^N a_{ij}(\omega_j - \omega_i) \left(\gamma^\omega \operatorname{sign} \left(\sum_{j \in \mathcal{N}_i} a_{ij}(\omega_j - \omega_i) \right) \right) \\ & + \eta^\omega \operatorname{sign}(\omega_i - \omega_{nom}) \sum_{i=1}^N \sum_{j=1}^N a_{ij}(\omega_j - \omega_i) + \beta^\omega \sum_{j \in \mathcal{N}_i} \mathcal{L}_A \omega_j \sum_{i=1}^N \sum_{j=1}^N a_{ij}(\omega_j - \omega_i) \\ & + \mathcal{V}_i^\omega \sum_{i=1}^N \sum_{j=1}^N a_{ij}(\omega_j - \omega_i) - g_1^\omega \left(\sum_{j \in \mathcal{N}_i} a_{ij}(\omega_j - \omega_i) \right)^{\frac{\mu_1}{\mu_2}} \sum_{i=1}^N \sum_{j=1}^N a_{ij}(\omega_j - \omega_i) \\ & - g_2^\omega \left(\sum_{j \in \mathcal{N}_i} a_{ij}(\omega_j - \omega_i) \right)^{\frac{\mu_3}{\mu_4}} \sum_{i=1}^N \sum_{j=1}^N a_{ij}(\omega_j - \omega_i)\end{aligned}\quad (32)$$

The above equation can be modified as follows:

$$\begin{aligned}\dot{V}(\omega) = & - \gamma^\omega \sum_{i=1}^N \left| \sum_{j=1}^N a_{ij}(\omega_j - \omega_i) \right| + \eta^\omega \operatorname{sign}(\omega_i - \omega_{nom}) \sum_{i=1}^N \sum_{j=1}^N a_{ij}(\omega_j - \omega_i) \\ & + \sum_{i=1}^N \sum_{j=1}^N a_{ij}(\omega_j - \omega_i) \left(\beta^\omega \sum_{j \in \mathcal{N}_i} \mathcal{L}_A \omega_j + \mathcal{V}_i^\omega \right) \\ & - g_1^\omega \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij}(\omega_j - \omega_i) \right)^{\frac{\mu_1 + \mu_2}{\mu_2}} \\ & - g_2^\omega \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij}(\omega_j - \omega_i) \right)^{\frac{\mu_3 + \mu_4}{\mu_4}}\end{aligned}\quad (33)$$

Define $\delta_{\omega_j, j \in \mathcal{N}_i} = \beta^\omega \sum_{j \in \mathcal{N}_i} \mathcal{L}_A \omega_j + \mathcal{V}_i^\omega$

So,

$$\begin{aligned}\dot{V}(\omega) = & - \gamma^\omega \sum_{i=1}^N \left| \sum_{j=1}^N a_{ij}(\omega_j - \omega_i) \right| + \sum_{i=1}^N \sum_{j=1}^N a_{ij}(\omega_j - \omega_i) \left(\eta^\omega \operatorname{sign}(\omega_i - \omega_{nom}) + (\delta_{\omega_j}) \right) \\ & - g_1^\omega \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij}(\omega_j - \omega_i) \right)^{\frac{\mu_1 + \mu_2}{\mu_2}} - g_2^\omega \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij}(\omega_j - \omega_i) \right)^{\frac{\mu_3 + \mu_4}{\mu_4}}\end{aligned}\quad (34)$$

Define $d_\omega = \max\{\eta^\omega + \delta_{\omega_1}, \dots, \eta^\omega + \delta_{\omega_N}\}$

$$\begin{aligned}\dot{V}(\omega) = & - (\gamma^\omega - d_\omega) \sum_{i=1}^N \left| \sum_{j=1}^N a_{ij}(\omega_j - \omega_i) \right| - g_1^\omega \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij}(\omega_j - \omega_i) \right)^{\frac{\mu_1 + \mu_2}{\mu_2}} \\ & - g_2^\omega \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij}(\omega_j - \omega_i) \right)^{\frac{\mu_3 + \mu_4}{\mu_4}}\end{aligned}\quad (35)$$

If γ^ω is chosen to be larger than d_ω , and using Lemmas 1 and 2, we can rewrite Equation (35) as

$$\dot{V}(\omega) \leq -g_1^\omega N^{\frac{\mu_2 - \mu_1}{2\mu_2}} \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij}(\omega_j - \omega_i)^2 \right)^{\frac{\mu_1 + \mu_2}{2\mu_2}} - g_2^\omega \sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij}(\omega_j - \omega_i)^2 \right)^{\frac{\mu_3 + \mu_4}{2\mu_4}}\quad (36)$$

Using Lemma 3 for $V(\omega) \neq 0$,

$$\sum_{i=1}^N \left(\sum_{j \in \mathcal{N}_i} a_{ij} (\omega_j - \omega_i) \right)^2 \geq \lambda_2(\mathcal{L}_G) \omega^T \mathcal{L}_G \omega \quad (37)$$

Using (37) in (36),

$$\dot{V}(\omega) \leq -g_1^\omega N^{\frac{\mu_2 - \mu_1}{2\mu_2}} (2|\lambda_2(\mathcal{L}_G)|V(\omega))^{\frac{\mu_1 + \mu_2}{2\mu_2}} - g_2^\omega (2|\lambda_2(\mathcal{L}_G)|V(\omega))^{\frac{\mu_3 + \mu_4}{2\mu_4}} \quad (38)$$

or

$$\dot{V}(\omega) \leq -g_1^\omega N^{\frac{\mu_2 - \mu_1}{2\mu_2}} y^{\frac{\mu_1 + \mu_2}{\mu_2}} - g_2^\omega y^{\frac{\mu_3 + \mu_4}{\mu_4}} \quad (39)$$

where $y = \sqrt{2|\lambda_2(\mathcal{L}_G)|V(\omega)}$. Take derivative of y :

$$\dot{y} = \frac{2|\lambda_2(\mathcal{L}_G)|\dot{V}(\omega)}{2\sqrt{2|\lambda_2(\mathcal{L}_G)|V(\omega)}} = \frac{\lambda_2(\mathcal{L}_G)\dot{V}(\omega)}{y} \quad (40)$$

Substitute $\dot{V}(\omega)$ from (40), we obtain

$$\dot{y} = |\lambda_2(\mathcal{L}_G)| \left(-g_1^\omega N^{\frac{\mu_2 - \mu_1}{2\mu_2}} y^{\frac{\mu_1}{\mu_2}} - g_2^\omega y^{\frac{\mu_3}{\mu_4}} \right) \quad (41)$$

or

$$\dot{y} \leq -ay^{\frac{\mu_1}{\mu_2}} - by^{\frac{\mu_3}{\mu_4}} \quad (42)$$

where $a = |\lambda_2(\mathcal{L}_G)|g_1^\omega N^{\frac{\mu_2 - \mu_1}{2\mu_2}}$ and $b = |\lambda_2(\mathcal{L}_G)|g_2^\omega$. Using Lemma 4.1 in [24], (42) represents a global finite-time stable system, where the convergence time is given as follows

$$T_\omega \leq \frac{1}{|\lambda_2(\mathcal{L}_G)|} \left(\frac{N^{\frac{\mu_1 - \mu_2}{2\mu_2}} \mu_2}{g_1^\omega (\mu_1 - \mu_2)} + \frac{\mu_4}{g_2^\omega (\mu_4 - \mu_3)} \right) \quad (43)$$

Next, we take the design of u_i^P as follows:

$$u_i^P = -c_P e_{Pi}^\omega = k_{Pi}^\omega \dot{P}_i \quad (44)$$

where e_{Qi}^ω is the neighborhood tracking error for reactive power sharing and can be given as

$$e_{Pi}^\omega = \sum_{j \in \mathcal{N}_i} a_{ij} (k_{Pi}^\omega P_i - k_{Pj}^\omega P_j) \quad (45)$$

The system can be rewritten in matrix form as follows:

$$k_P^\omega \dot{P} = u^P \quad (46)$$

$$u^P = -c_P e_P^\omega \quad (47)$$

$$e_P^\omega = \mathcal{L}_G k_P^\omega P \quad (48)$$

where $k_P^{\omega} \dot{P} = [k_{P1}^{\omega} \dot{P}_1, k_{P2}^{\omega} \dot{P}_2, \dots, k_{PN}^{\omega} \dot{P}_N]^T$, $u^P = [u_1^P, u_2^P, \dots, u_N^P]^T$, $e_P^{\omega} = [e_{P1}^{\omega}, e_{P2}^{\omega}, \dots, e_{PN}^{\omega}]^T$, $k_P^{\omega} P = [k_{P1}^{\omega} P_1, k_{P2}^{\omega} P_2, \dots, k_{PN}^{\omega} P_N]^T$. Now, consider a Lyapunov function,

$$V(P) = \frac{1}{2} e_P^{\omega T} R e_P^{\omega} \quad (49)$$

With $R = R^T$ and $R > 0$ is a positive semidefinite matrix. The derivative of Lyapunov function is as follows:

$$\dot{V}(P) = e_P^{\omega T} R L_G u^P \quad (50)$$

Define $M \equiv \mathcal{L}_G$ using (47) in (50),

$$\dot{V}(P) = -c_P e_P^{\omega T} R M e_P^{\omega} = -\frac{1}{2} c_P e_P^{\omega T} (R M + M^T R) e_P^{\omega} \quad (51)$$

From [25], $R M + M^T R$ is a positive definite matrix, there $\dot{V}(P)$ will be negative; thus, the proposed controller is asymptotically stable. The controller in Equation (44) will accurately share the reactive power in a longer time frame when the secondary control is activated. $\square$

Figure 3 shows the block diagram for proposed frequency and active power-sharing algorithm.

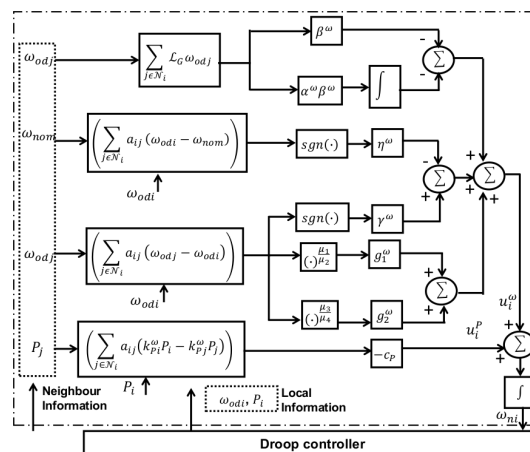


Figure 3. Block diagram of proposed secondary frequency and active power regulation scheme.

4. Results and Discussion

An islanded AC test MG network is shown in Figure 4a and is simulated in the MATLAB/ SimPowerSystem environment. The reference voltage and frequency are $V_{ref} = 415$ V, $f_{ref} = 50$ Hz, respectively. The considered test system is composed of five DG units and their respective loads. The loads considered are the series R-L loads and the transmission lines are modeled as lumped R-L. The detailed system parameters, line impedances, and loads are summarized in Table 1. The loads connected to the MG network are the R-L loads, where Load 1 to Load 5 are given as follows: $R = 300 \Omega$ - $L = 477$ mH, $R = 40 \Omega$ - $L = 64$ mH, $R = 50 \Omega$ - $L = 64$ mH, $R = 40 \Omega$ - $L = 64$ mH, and $R = 50 \Omega$ - $L = 95$ mH. Table 2 gives the parameters for the proposed algorithm. The DG units share information via a sparse communication network with a weighted graph, as shown in Figure 4b. The Laplacian matrix ($\mathcal{L}_G$) for the undirected weighted graph given in Figure 4b is written as follows:

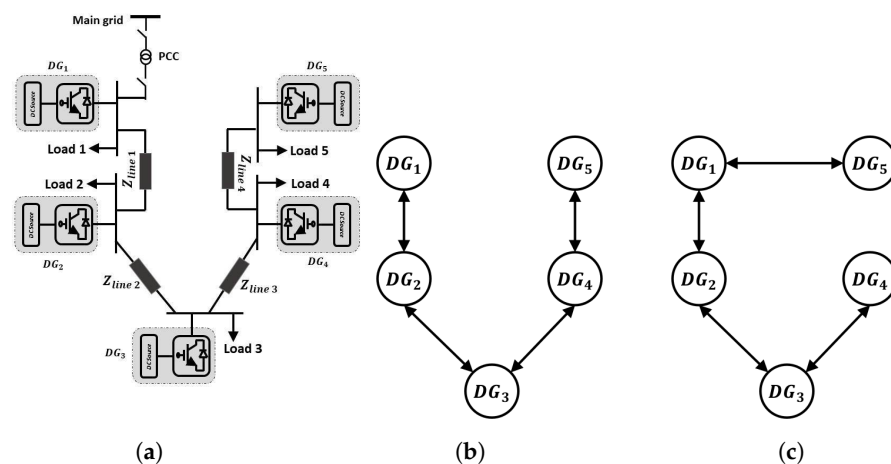
$$\mathcal{L}_G = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix} \quad (52)$$

Table 1. Parameters of MG Test System.

	DG 1–3		DG 4 and 5	
DGs	k_Q^V	9.4×10^{-5}	k_Q^V	12.5×10^{-5}
	k_P^ω	1.3×10^{-3}	k_P^ω	1.5×10^{-3}
	R	0.03Ω	R	0.03Ω
	L	0.35 mH	L	0.35 mH
	R_f	0.1Ω	R_f	0.1Ω
	L_f	1.35 mH	L_f	1.35 mH
	C_f	$50 \mu\text{F}$	C_f	$50 \mu\text{F}$
	K_{PV}	0.1	K_{PV}	0.05
	K_{IV}	420	K_{IV}	390
	K_{PC}	15	K_{PC}	10.5
	K_{IC}	$20,000$	K_{IC}	$16,000$
Lines	$Z_{Line 12}$	$Z_{Line 23}$	$Z_{Line 34}$	$Z_{Line 45}$
	$R_l = 0.12 \Omega$	$R_l = 0.175 \Omega$	$R_l = 0.12 \Omega$	$R_l = 0.175 \Omega$
	$L_l = 0.318 \text{ mH}$	$L_l = 1.847 \text{ mH}$	$L_l = 0.318 \text{ mH}$	$L_l = 1.847 \text{ mH}$

Table 2. Controller gains of proposed algorithm.

$\gamma^V = 20$	$\alpha^V = 3$	$\beta^V = 10$	$c_P = 7$
$\gamma^\omega = 20$	$\alpha^\omega = 3$	$\beta^\omega = 10$	$c_Q = 10$
$\mu_1 = 5$	$\mu_2 = 3$	$\mu_3 = 5$	$\mu_4 = 7$
Case 1		$g_1^V = 80, g_2^V = 60$	$g_1^\omega = 80, g_2^\omega = 60$
Case 2		$g_1^V = 40, g_2^V = 30$	$g_1^\omega = 40, g_2^\omega = 30$

**Figure 4.** (a) A schematic diagram of the studied AC microgrid test system. (b) Communication graph for Section 4.1. (c) Communication graph for Section 4.4.

To verify the performance of the presented controllers, initially only the primary control level is working, which leads to the voltage and frequency perturbation from their reference values. The simulation is performed for different scenarios as follows.

4.1. Performance Evaluation under Load Variation

To demonstrate the robustness of the proposed control technique, the inclusion and removal of the load are considered. The communication topology used is shown in Figure 4b.

The secondary control parameters are given in case 1, Table 2. The simulation is pursued in four steps:

Step 1: At $t = 0$ s, the primary droop control is activated exclusively.

Step 2: At $t = 2$ s, the presented restoration algorithms in (3) and (25) are activated.

Step 3: At $t = 3$ s, a new load is added to load 1.

Step 4: At $t = 4$ s, load 3 is removed.

During stage 1 of the operation, the primary control layer is active, which results in voltage and frequency deviations from their reference values, as shown in Figure 5. During the second step of the operation, at $t = 2$ s, the proposed secondary restoration algorithm is activated and the voltage and frequency for each DG unit are restored to their nominal values ($V_{nom} = 415$ V and $\omega_{nom} = 50$ Hz) in finite-time $t \leq 0.25$ s. At $t = 3$ s, an RL load of $R = 30 \Omega$, $L = 47$ mH is added in load 1, and at $t = 4$ s, load 3 is removed from the MG network. The presented scheme is effective in rejecting the load changes and tight voltage, and the frequency restoration is observed with a small transient time. Active power is shared in accordance with the DG ratings, as shown in Figure 5c, and the reactive power results show a small mismatch because of the MG line impedance, as shown in Figure 5d, and the power-sharing works in a longer time frame. Thus, the proposed restoration algorithm is effective in rejecting the load variation in the MG.

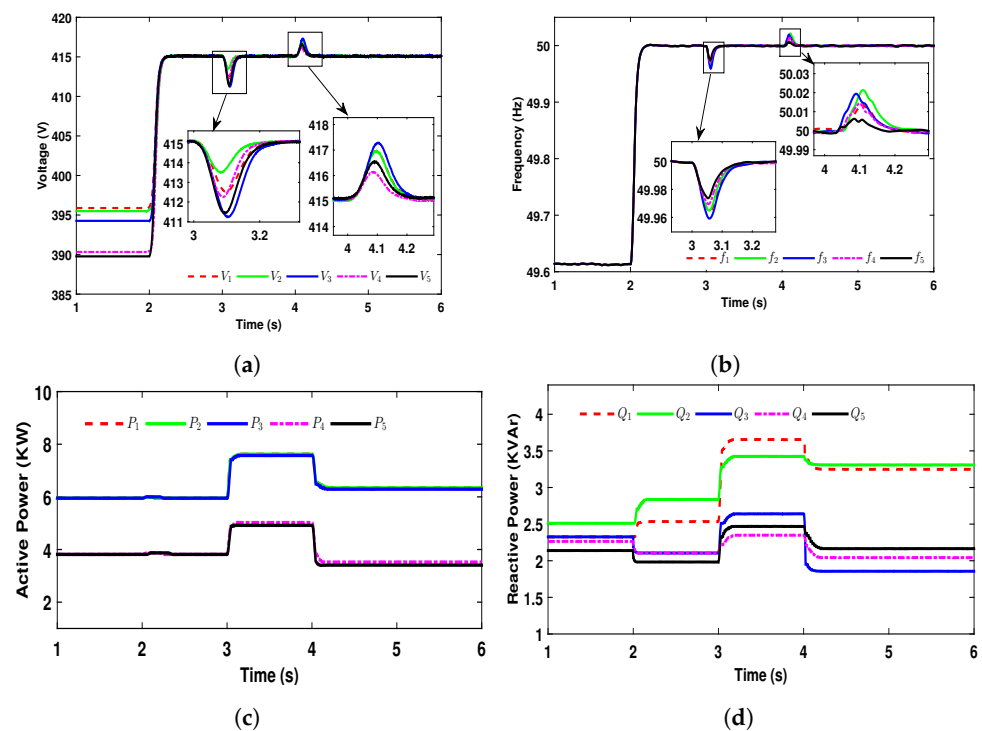


Figure 5. Performance analysis of proposed controller. (a) Voltage. (b) Frequency. (c) Active power. (d) Reactive power.

4.2. Analysis of Convergence Speed

The convergence speed can be tuned based on the user's requirement by adjusting the gain parameters in the proposed finite-time algorithm. By setting gains g_1^V and g_2^V as per Section 4.1, the upper limit of the convergence time given by (18) can be calculated as $T_V \leq \frac{1}{0.382} \left(\frac{5^2 \times 3}{80 \times (5-3)} + \frac{7}{60 \times (7-5)} \right) = 0.236$ s. In the simulation, the convergence time is approximately 0.2s, as shown in Figure 5. Thus, the convergence time closely matches with the numerical value $T_V \leq 0.236$ s. The convergence speed can be manipulated with ease via tuning the gain values as half of the previous values, as shown in Table 2, case 2. The simulation results in Figure 6 clearly show that the convergence time is approximately 0.4 s. By using (18), the upper bound of T_V can be calculated as $T_V \leq$

$\frac{1}{0.382} \left(\frac{\frac{5-3}{52 \times 3} \times 3}{40 \times (5-3)} + \frac{7}{30 \times (7-5)} \right) = 0.47$ s which validates the correctness of the upper bound of the restoration time given in (18). Though the convergence time is independent of the load and line and DG parameters in the MG, it should be noted from (18) that T_V relies on the communication connectivity of the MG network, and therefore a change in the topology will affect the convergence time accordingly.

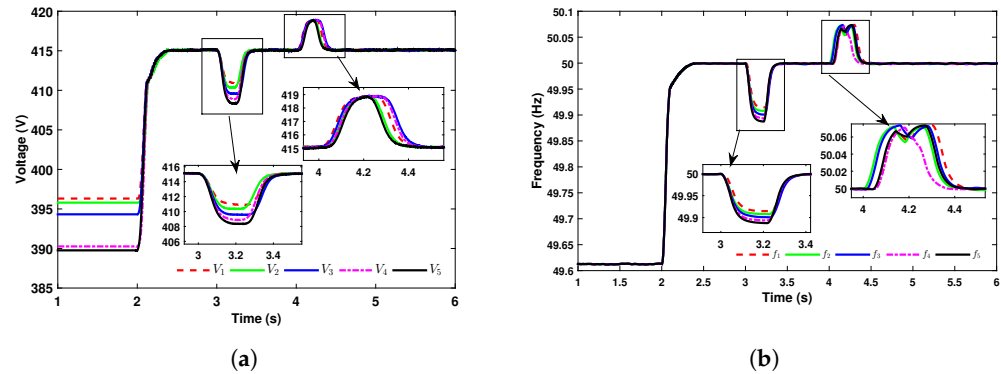


Figure 6. Convergence performance of the proposed restoration algorithm. (a) Voltage. (b) Frequency.

4.3. Plug-and-Play Capability

The efficacy of the presented finite-time controller is tested for the plug-and-play (PNP) feature of the MG, as the renewable generation is intermittent in nature. The continuous connection and disconnection of the DGs to the MG network require a seamless synchronization of the voltage and frequency which is necessary along with accurate power sharing among DGs. Losing one DG unit causes the change in the communication network topology, as the communication link of that DG also fails. To demonstrate the efficacy, DG 5 is disconnected from the MG to mimic the loss of a DG unit. The loss of DG 5 causes the failure of links 4–5, but the other links still make a connected graph. Figure 7 shows the performance of the presented controller under the PNP operation. As DG 5 is disconnected, the other DG units in the MG generate adequate power to compensate for the loss. The power output of the disconnected DG becomes zero with a slower inclination as the low-pass filter provides harmonic rejection. The voltage and frequency regulation is maintained with a small transient time. At $t = 4$ s, DG 5 is connected back to the MG. The results show the effective voltage and frequency regulation with proper active power sharing among the DGs. Hence, the proposed scheme supports the PNP operation.

4.4. Performance with Communication Topology Change

The efficacy of the proposed finite-time secondary controller is verified with a different communication topology and is demonstrated in this case. The communication link between DG 4 and DG 5 does not exist, as shown in Figure 4c. The change in the communication topology does not affect the controller performance. All the other nodes remain connected to receive the reference values of the voltage and frequency from DG 1. The results shown in Figure 8 clearly depict that for all DG units, the voltage and frequency are synchronized back to their nominal values in definite time $t < 0.25$ s, and the active power is shared according to their ratings.

4.5. Performance with Data Dropout

The data dropout or packet loss that occurs in the communication system negatively affects the efficacy of the controller. The packet loss means the loss of the measured information of the neighboring DG unit (the voltage magnitude, operating frequency, and real and reactive power outputs) is lost. The excessive packet loss leads to significant variations in the voltage and frequency terms. Here, two situations of data dropout are simulated. In the first case, the data dropout takes place every 30 ms in all the links, and in the second case, it is every 10 ms in all the links considering 100 ms of communication

delay. The proposed synchronization algorithm remains robust for the 1st case, as shown in Figure 9a,b, and for the 2nd case, the response is not significantly affected, as seen from Figure 9c,d.

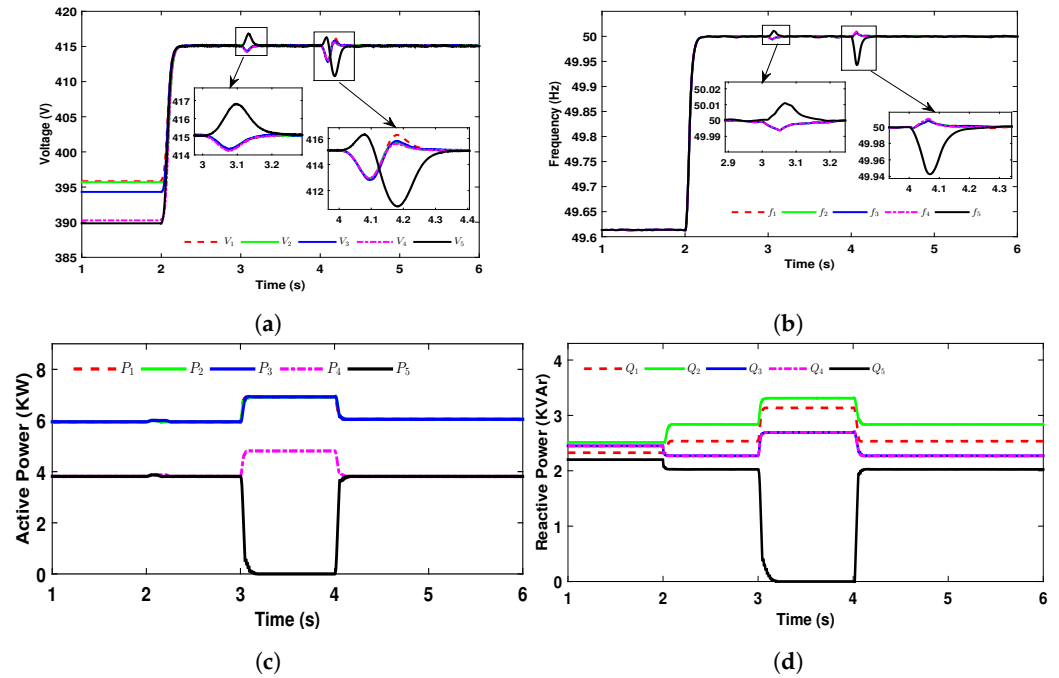


Figure 7. Performance with loss of DG unit (PNP). (a) Voltage. (b) Frequency. (c) Active power. (d) Reactive power.

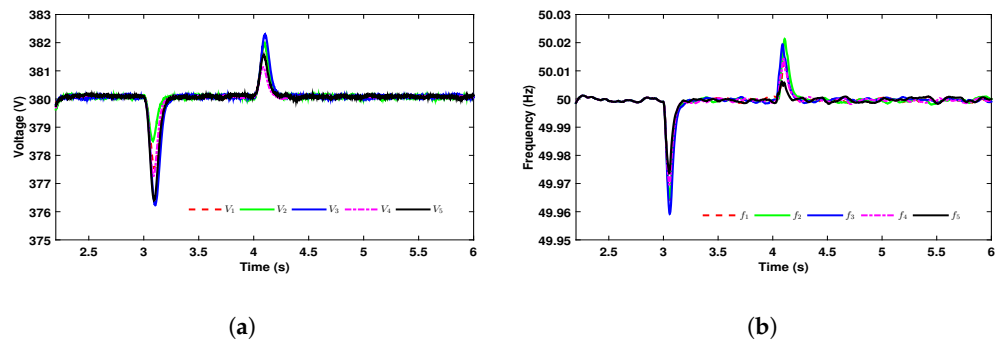


Figure 8. Restoration with communication topology change. (a) Voltage. (b) Frequency.

4.6. Performance with Communication Link Latency

The communication links have an inherent link latency due to the data traffic and distance between the DGs, which is usually not larger than 100 ms. This adversely affects the controller efficacy and may result in system instability for larger link delays [26]. The link delay analysis is presented in this case. The measured information of the neighboring DG unit, i.e., the voltage magnitude, operating frequency, and active and reactive power outputs, is delayed which further results in an increased convergence time. For this analysis, we considered three fixed link delays as (τ_d) of 100 ms, 1 s, and 2 s. The proposed controller has an insignificant effect for the delays $\tau_d = 100$ ms and $\tau_d = 1$ s, as seen from Figure 10. For longer link delays, the proposed controller synchronizes the voltage and frequency with a slightly longer convergence time with moderate ripples. For the communication delay $\tau_d = 1$ s and $\tau_d = 2$ s, the proposed scheme restores the voltage and frequency in approximately 0.31 and 0.42 s, respectively, as shown in Figure 11.

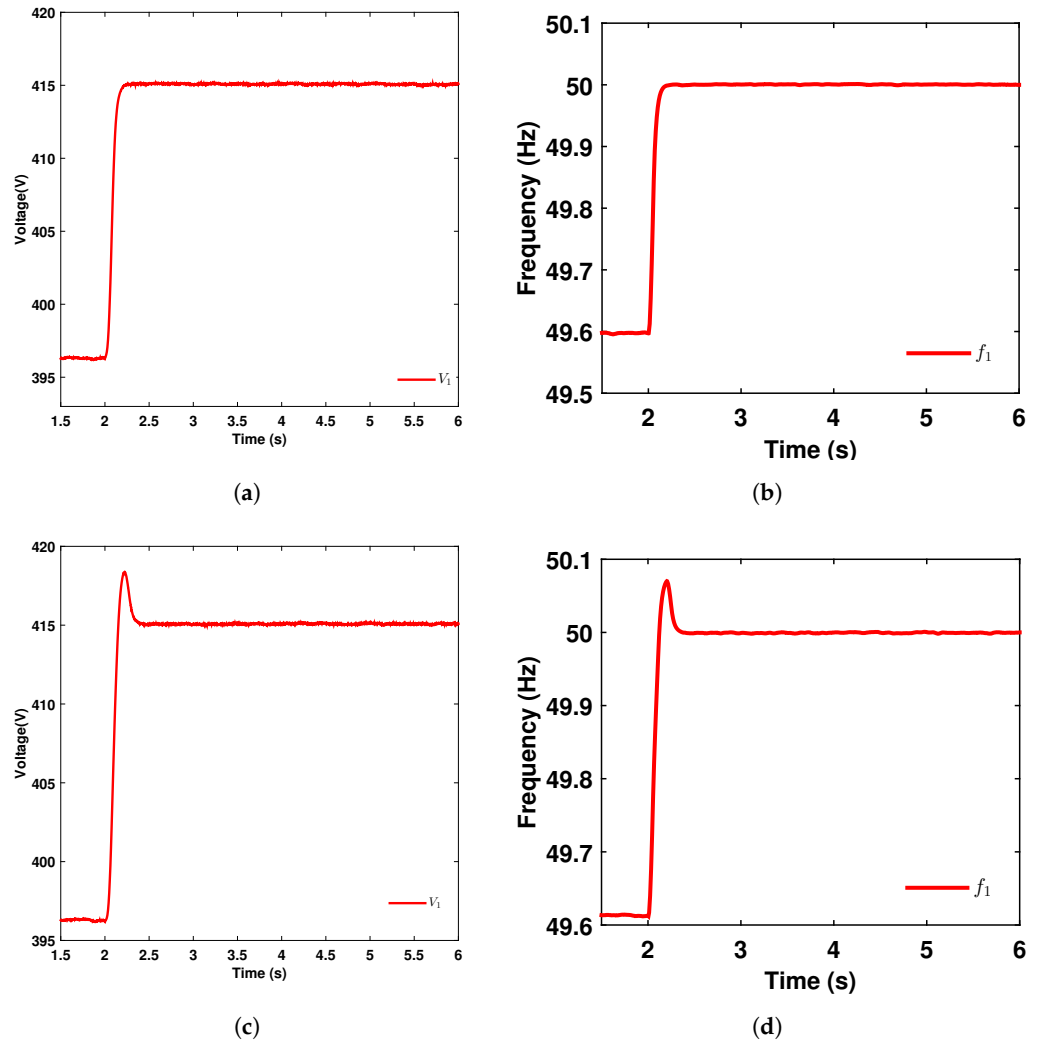


Figure 9. Performance with data dropout. (a) Voltage every 30 ms. (b) Frequency every 30 ms. (c) Voltage every 10 ms. (d) Frequency every 10 ms.

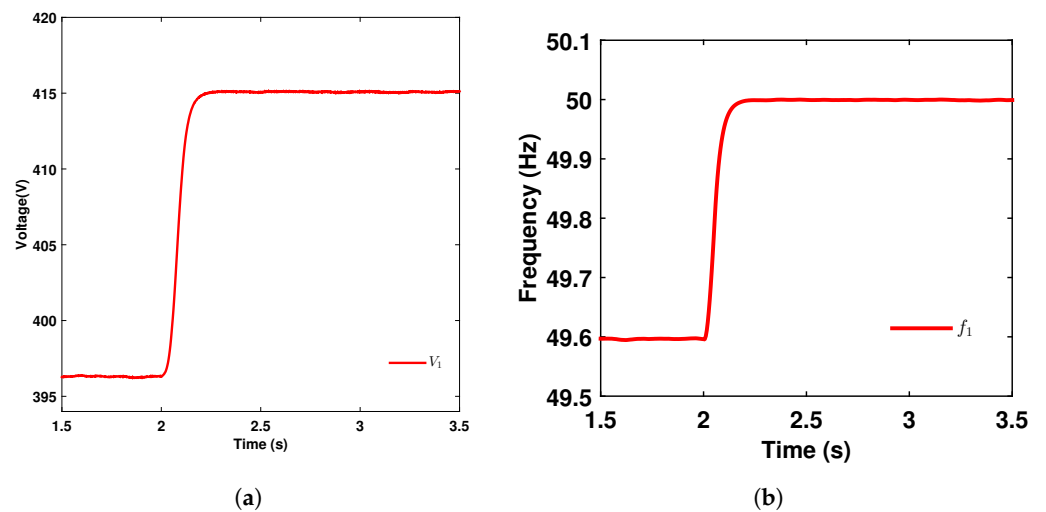


Figure 10. Performance with link delay. (a) Voltage under link delay $\tau_d = 100$ ms. (b) Frequency under link delay $\tau_d = 100$ ms.

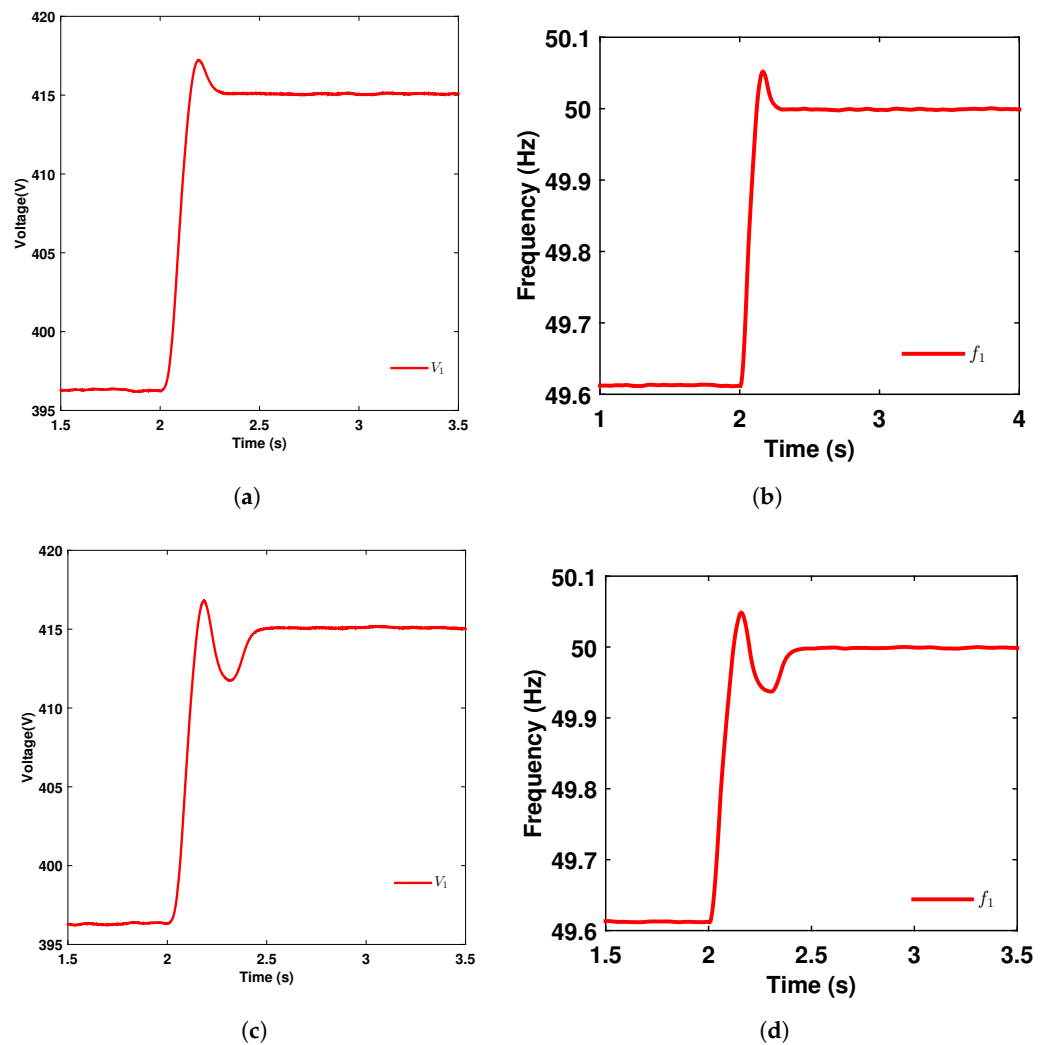


Figure 11. Performance under link delay. (a) Voltage under link delay $\tau_d = 1$ s. (b) Frequency under link delay $\tau_d = 1$ s. (c) Voltage under link delay $\tau_d = 2$ s. (d) Frequency under link delay $\tau_d = 2$ s.

4.7. Comparative Result Analysis with Communication Network Impairments

As stated, a few works consider both the voltage and frequency synchronization in the finite time together [5]. To show the effectiveness of the proposed controller in (3) and (25) over the existing ones, a comparative evaluation was performed with the asymptotic controller presented in [27] under adverse communication conditions. The test setup and the MG's parameters remain similar for both the controllers under consideration for comparison. The controllers are implemented with the loss of a data packet every 10 ms in all the communication links with a delay of 100 ms to replicate an adverse communication channel with losses and link delays. The proposed controller restores the voltage and frequency in less than 0.25 s, whereas the asymptotic controller takes approximately 0.5 s for the synchronization. Hence, the proposed controller outperformed the asymptotic controller in [27] in terms of the restoration time by 54%. The performance improvement is clearly visible in Figure 12. The intermittent delay and data loss affect the asymptotic controller performance and it becomes unstable, whereas the proposed finite-time controller restores the voltage and frequency within a permissible time.

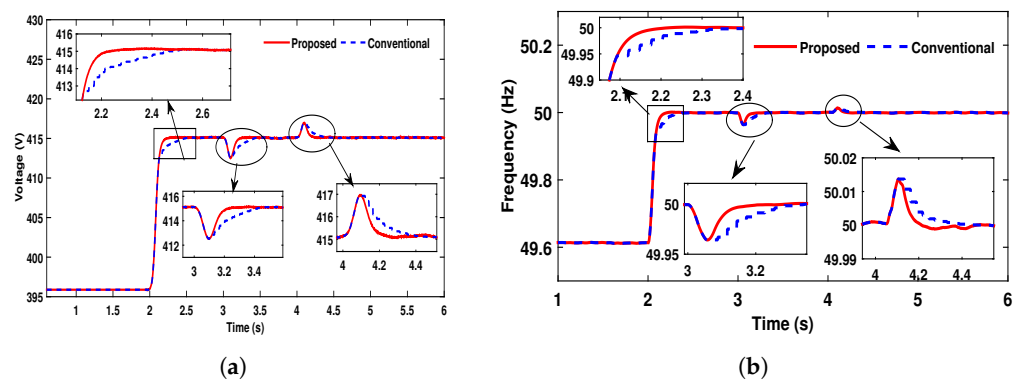


Figure 12. Comparative analysis. (a) Voltage magnitude. (b) Frequency.

5. Conclusions

A two-layer hierarchical cyber-physical framework is introduced for inverter-based DGs in an islanded AC MG. The cyber layer is augmented with an MG structure for information sharing over a sparse communication network. The proposed controller is distributed; hence, the central controller is not required. This minimizes the computation and risk of one-point failure. The proposed control scheme synchronizes the operating voltage and frequency in a definite convergence time. The Lyapunov analysis is pursued on the upper limit of the restoration time. The proposed controller has the flexibility in the regulation of the convergence time to meet specific user demands and different operating conditions. Further, the proposed asymptotic power-sharing scheme enables the correct active power regulation in a longer time frame. The efficacy of the proposed controller was demonstrated via the time-domain simulation of the MG test system in MATLAB. The obtained results display that the proposed controller provides a provision for ancillary services under the event of load variations and information network constraints, such as a time delay, data dropout, and topology change. The proposed scheme also meets the PNP requirement of the AC MG and outperforms the existing neighborhood tracking error-based asymptotic controllers. Accurate reactive power sharing, an analysis of the computational complexity, and protection under cyber threats are the limitations of this work the author will address in future research work.

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